

Linyi (Tiger) Hou

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EDUCATION

Ph.D. in Aerospace Engineering

May 2025

University of Illinois Urbana-Champaign

GPA: 3.94/4.00

Thesis: Autonomous Lost in Space and Time State Determination using Periodic Variable Stars

B.S. in Aerospace Engineering, Minor in Computer Engineering

May 2021

University of Illinois Urbana-Champaign

GPA: 3.93/4.00

SKILLS

Technical batch filtering, Kalman filtering, Monte Carlo simulation, celestial navigation, orbit determination, spacecraft navigation modeling

Collaboration technical/proposal writing, MS Office suite, $\text{\LaTeX}$, Git, GitHub, Trello

Programming Python, C/C++, MATLAB/Simulink, GMAT, Javascript, Julia, ADQL

RESEARCH

Astrodynamics and Planetary Exploration Group (APEX)

Advisor: Dr. Siegfried Eggel

Interstellar Navigation and Star Identification

Sep 2024

- Developed a star identification technique using C++ which mitigates stellar aberration effects at relativistic speeds, improving star identification success rate from 38% to 91%. – May 2025

Celestial Navigation to Supplement DSN Tracking

Jul 2024

- Collaborated with colleagues to create navigation simulation tools in Python to assess the feasibility of reducing Deep Space Network tracking cadence by using onboard celestial navigation for the OSIRIS-APEX and CAPSTONE missions. – Mar 2025
- Designed sensor scheduling algorithms for celestial navigation which reduced navigation uncertainty by over 10% compared to existing strategies in Monte Carlo simulation.
- Implemented spherical harmonics for an open source orbit propagator in C++.

Lost in Space and Time Navigation using Variable Stars

Nov 2022

- Developed an autonomous optical navigation concept using variable star light curves which delivers timing accuracy within 1 second without needing prior position/time information. – Mar 2025
- Simulated star tracker observations and demonstrated the feasibility of reestablishing Earth pointing via the proposed method through Monte Carlo analysis.

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Aerospace Mission Analysis Laboratory

Advisor: Dr. Zachary R. Putnam

Lost in Space X-ray Pulsar Navigation (XNAV)

Jun 2021

- Developed algorithms to resolve spacecraft position and time from pulsar timing data.
- Created a photon time of arrival simulation tool in Python and validated simulation results against the pulsar timing package PINT to within 50 μ s.

– Dec 2023

Velocity-based Initial Orbit Determination

Aug 2019

- Developed a novel initial orbit determination technique using batched sequential range-rate measurements that achieved over 10x better accuracy than state-of-the-art methods.

– Apr 2024

Space Systems Optimization Laboratory

Advisor: Dr. Koki Ho

Architecture Trade Studies on ISRU Technologies for Human Space Exploration

Sep 2018

- Worked with two graduate students to discuss the development of a mixed-integer linear programming framework for space logistics optimization.
- Conducted literature review of in-situ resource utilization (ISRU) technologies to quantify input/output rates of technology options for an architecture optimization tool.

– Sep 2019

WORK EXPERIENCE

Teaching Assistant

Jun 2021

University of Illinois Urbana-Champaign (AE 202, AE 443, AE 483)

– May 2024

- Worked with 2-5 faculty and staff to teach lessons, hold office hours, and improve curriculum for courses in flight mechanics, systems engineering, and UAV controls.
- Helped 20+ students formulate and implement UAV state-space model controllers with various sensors, e.g., laser rangefinders, optical flow cameras, and time-of-flight anchors.
- Assisted in developing LQR control-related code and instructional materials for a UAV course using C and Python.

Software Engineering Intern

May 2019

Collins Aerospace

– Aug 2019

- Migrated an avionics data interpreter from 16-bit to 32-bit using C++ and Python.
- Automated weekly builds and integration in TortoiseSVN using Python.

PROJECTS

Collaborative Projects in Flight Dynamics	2022, 2023
- Planetary entry: Created a 3-DOF entry simulation to assess the flight performance of several novel Apollo-derived entry guidance algorithms at Mars. - Trajectory optimization: Applied optimal control theory to identify time- and fuel-optimal low-thrust Earth-Mars transfer trajectories.	
Team Lead/Advisor, NASA's RASC-AL Competition	2019, 2022, 2023
- Led a team of 10+ students to develop a reusable crewed lunar lander concept (structure, power, thermal, life support, orbital mechanics, budget, etc.). - Advised two student teams who developed technical requirements and performed trade studies for lunar and Martian in-situ resource utilization architectures.	

HONORS AND AWARDS

Aerospace Engineering Alumni Advisory Board Fellowship	2024
<i>Department of Aerospace Engineering, University of Illinois Urbana-Champaign</i>	
Best Student Paper, 46 th AAS GN&C Conference	2024
<i>American Astronautical Society (AAS) Rocky Mountain Section</i>	
Yee Fellowship	2021, 2024
<i>Grainger College of Engineering, University of Illinois Urbana-Champaign</i>	
David and Catherine Thompson Space Technology Scholarship	2020
<i>American Institute of Aeronautics and Astronautics (AIAA)</i>	
H.S. Stillwell Problem Solving Award	2020
<i>Department of Aerospace Engineering, University of Illinois Urbana-Champaign</i>	
Robert W. McCloy Memorial Award	2019
<i>Department of Aerospace Engineering, University of Illinois Urbana-Champaign</i>	
James Scholar	2018 - 2021
<i>University of Illinois Urbana-Champaign</i>	
Dean's List	2017 - 2021
<i>University of Illinois Urbana-Champaign</i>	

JOURNAL ARTICLES

- [J.5] **L. Hou**, Ishaan Bansal, Clark Davis, and Siegfried Eggl. “Position and Time Determination without Prior State Knowledge via Onboard Optical Observations of Delta Scuti Variable Stars”. In: *The Journal of the Astronautical Sciences* 72.9 (2025). DOI: [10.1007/s40295-025-00485-8](https://doi.org/10.1007/s40295-025-00485-8).
- [J.4] **L. Hou** and Zachary R Putnam. “Autonomous Initial Orbit Determination using Sequential Range-Rate Measurements”. In: *The Journal of the Astronautical Sciences* 71.24 (2024). DOI: [10.1007/s40295-024-00442-x](https://doi.org/10.1007/s40295-024-00442-x).
- [J.3] **L. Hou** and Zachary R Putnam. “A Norm-Minimization Algorithm for Solving the Lost-in-Space Problem with XNAV”. In: *The Journal of the Astronautical Sciences* 71.6 (2024). DOI: [10.1007/s40295-023-00425-4](https://doi.org/10.1007/s40295-023-00425-4).
- [J.2] Hao Chen, Tristan Sarton du Jonchay, **L. Hou**, and Koki Ho. “Multifidelity Space Mission Planning and Infrastructure Design Framework for Space Resource Logistics”. In: *Journal of Spacecraft and Rockets* 58.2 (2021), pp. 538–551. DOI: [10.2514/1.A34666](https://doi.org/10.2514/1.A34666).
- [J.1] Hao Chen, Tristan Sarton du Jonchay, **L. Hou**, and Koki Ho. “Integrated In-Situ Resource Utilization System Design and Logistics for Mars Exploration”. In: *Acta Astronautica* 170 (2020), pp. 80–92. DOI: [10.1016/j.actaastro.2020.01.031](https://doi.org/10.1016/j.actaastro.2020.01.031).

CONFERENCE PROCEEDINGS

- [C.8] **L. Hou** and Siegfried Eggl. “Autonomous Optical Navigation using Astrometry and Photometry”. In: *AAS/AIAA Space Flight Mechanics Meeting*. AAS 25-248. Kaua’i, HI, Jan. 2025.
- [C.7] **L. Hou**, Ishaan Bansal, Clark Davis, and Siegfried Eggl. “Lost-in-space Position and Time Determination via Star Tracker Observations of Periodic Variable Stars”. In: *46th Rocky Mountain AAS GN&C Conference*. AAS-24-012. Breckenridge, CO, Feb. 2024.
- [C.6] Alec Auster, Ana Bojinov, Shikhar Kesarwani, Galen Sieck, Riya Shah, Madeline Odeen, Sara McCarthy, Madison Frankenthor, Komol Patel, Grant Davis, et al. “Mars Ice Thermal Harvesting Rig & ISRU Laboratory (MITHRIL)”. In: *ASCEND 2022*. Aug. 2022, p. 4249. DOI: [10.2514/6.2022-4249](https://doi.org/10.2514/6.2022-4249).
- [C.5] **L. Hou** and Zachary R. Putnam. “A Norm-minimization Method for Solving the Cold-Start Problem with XNAV”. In: *AAS/AIAA Astrodynamics Specialist Conference*. AAS 22-560. Charlotte, NC, Aug. 2022.
- [C.4] **L. Hou** and Zachary R. Putnam. “Initial Orbit Determination from Sequential Line-of-Sight Velocity Measurements”. In: *AAS/AIAA Astrodynamics Specialist Conference*. AAS 22-561. Charlotte, NC, Aug. 2022.

- [C.3] **L. Hou**, Kevin Lohan, and Zachary R Putnam. “Comparison and Error Modeling of Velocity-Based Initial Orbit Determination Algorithms”. In: *AAS/AIAA Space Flight Mechanics Meeting*. AAS 21-280. Virtual, Feb. 2021.
- [C.2] Hao Chen, Tristan Sarton du Jonchay, **L. Hou**, and Koki Ho. “Multi-Fidelity Space Mission Planning and Space Infrastructure Design Framework for Space Resource Logistics”. In: *AIAA Propulsion and Energy 2019 Forum*. 2019, p. 4134. DOI: [10.2514/6.2019-4134](https://doi.org/10.2514/6.2019-4134).
- [C.1] H Chen, T Sarton du Jonchay, **L. Hou**, and K Ho. “Integrated Analysis Framework for Space Propellant Logistics: Production, Storage, and Transportation”. In: *Lunar ISRU 2019-Developing a New Space Economy Through Lunar Resources and Their Utilization* 2152 (2019), p. 5003. URL: <https://www.hou.usra.edu/meetings/lunarisru2019/pdf/5003.pdf>.